

1 The nilpotent filtration on every independent arithmetic face

This construction applies Deligne's nilpotent-filtration linear algebra to the actual arithmetic packets and to every independent coefficient face. It does not assert an arithmetic purity estimate. Every formula below retains the original arithmetic dilation, its scalar eigenvalue, and all its nilpotent terms. The additional grading operator is recorded as an additional operator.

1.1 Original packets and the filtration

Let Z be a finite set of distinct zeros of the original function $g(s) = 2\xi(s)$, with their full orders $m_\rho \geq 1$. Put

$$h_Z(x) = \prod_{\rho \in Z} (x - \rho)^{m_\rho}, \quad E_Z = \mathbb{C}[x]/(h_Z), \quad E_\rho = e_\rho E_Z. \quad (\text{MW1})$$

Here e_ρ is the Chinese-remainder idempotent, and $u_\rho = e_\rho(x - \rho)$ is an element of this algebra, with local unit e_ρ . Write $v_{\rho,r} = e_\rho(x - \rho)^r$, for $0 \leq r < m_\rho$. These vectors are the original local jet basis. The nilpotent operator N_ρ is multiplication by u_ρ :

$$N_\rho v_{\rho,r} = v_{\rho,r+1} \quad (r < m_\rho - 1), \quad N_\rho v_{\rho,m_\rho-1} = 0. \quad (\text{MW2})$$

For an integer j , define the increasing filtration

$$M_j E_\rho = \text{span}_{\mathbb{C}}\{v_{\rho,r} : m_\rho - 1 - 2r \leq j\}, \quad M_j E_Z = \bigoplus_{\rho \in Z} M_j E_\rho. \quad (\text{MW3})$$

In particular a zero vector in $M_j E_\rho$ remains the zero of its specified coefficient face when this filtration is split-lifted.

Filtration calculation. Multiplication by u_ρ lowers the displayed index by exactly two. Thus $N_\rho M_j \subseteq M_{j-2}$. For $k \geq 0$, the only way $\text{Gr}_k^M E_\rho$ is nonzero is

$$k = m_\rho - 1 - 2r \geq 0, \quad 0 \leq r < m_\rho.$$

It is then one dimensional with basis the class of $v_{\rho,r}$. The map induced by N_ρ^k sends this basis vector to the class of $v_{\rho,r+k}$, whose index is $-k$ and whose exponent $r+k = m_\rho - 1 - r$ lies between zero and $m_\rho - 1$. Its coefficient is exactly one. Thus

$$N_\rho^k : \text{Gr}_k^M E_\rho \xrightarrow{\sim} \text{Gr}_{-k}^M E_\rho. \quad (\text{MW4})$$

When the source graded space is zero, the target is zero by the same index calculation. Direct sums prove the corresponding assertion for $N = \bigoplus_\rho N_\rho$ on E_Z . This is precisely the centered nilpotent filtration characterized in Weil II, Proposition 1.6.1.

There is an intrinsic formula which also proves functoriality without choosing a new jet basis:

$$M_j V = \sum_{i \geq 0} \left(\ker N^{i+1} \cap \text{im } N^{\max(0, i-j)} \right), \quad V = E_Z. \quad (\text{MW5})$$

To verify it on a block of length m , both spaces in each intersection are terminal spans of the original basis. Their intersection starts at exponent $\max(0, m-1-i, i-j)$. The sum over i starts at the smallest such exponent, namely $\max(0, -j, \lceil (m-1-j)/2 \rceil)$; an exponent at least m denotes the zero space. When the extra term $-j$ exceeds the other terms, $j < 1-m$ and both spans are zero. Otherwise the starting exponents agree with MW3. Kernels and images commute with the finite direct-sum decomposition, proving MW5. A linear map commuting with the nilpotent operators carries each kernel and each image into the corresponding kernel and image. It therefore preserves MW5.

1.2 Independent masks and the arithmetic action

For $n \geq 1$, retain the independent carrier

$$D_{E_Z, n} = G(E_Z)^n, \quad G(E_Z) = \{\tau\} \sqcup \{b^\bullet : b \in E_Z\}. \quad (\text{MW6})$$

For each $A \subseteq I_n = \{0, \dots, n-1\}$, its face is $\{A\} \times E_Z^A$, with the empty face a single absent point. The filtration on the entire carrier is the collection

$$\widetilde{M}_j D_{E_Z, n} = \{\tau\} \sqcup \coprod_{\emptyset \neq A \subseteq I_n} \{A\} \times (M_j E_Z)^A. \quad (\text{MW7})$$

This notation specifies additive faces and their filtrations. No assertion that each $\widetilde{M}_j$ is a unital subsemiring is made. Every present face has its own zero $(A, 0)$ in every filtration step; the filtration does not turn any of these zeros into absence. The zero-insertion maps for $A \subseteq B$ preserve MW7 because they insert the original zero of $M_j E_Z$.

The original arithmetic operator, for real $a > 0$, is multiplication by the full jet of $a^s = \exp(s \log a)$. On a local block it is

$$A_a|_{E_\rho} = a^\rho \sum_{r=0}^{m_\rho-1} \frac{(\log a)^r}{r!} N_\rho^r. \quad (\text{MW8})$$

It preserves MW3, and its induced action on every nonzero graded piece of that block is exactly a^ρ , since all terms with $r > 0$ lower the filtration index. Its inverse is $A_{a^{-1}}$. Thus the split lift preserves every independent mask, including the zero in each active slot.

For an odd prime power q with $\gcd(q, n) = 1$, retain the coefficient Frobenius from the original negacyclic algebra:

$$\Psi_q(v t^i) = (-1)^{\lfloor qi/n \rfloor} v t^{qi \bmod n}, \quad B_n = E_Z[t]/(t^n + 1). \quad (\text{MW9})$$

It sends A to qA and leaves the jet index unchanged. Consequently it preserves the full filtered face diagram, with its specified permutation of face labels and its original signs. It commutes with N and A_a . Define the combined operator, at its full linear coefficient-space type, by

$$\begin{aligned} B_{\rho, n} &= E_\rho[t]/(t^n + 1), \quad M_j B_{\rho, n} = \bigoplus_{i=0}^{n-1} (M_j E_\rho) t^i, \\ T_{a, q} &= A_a \Psi_q = \Psi_q A_a : B_n \longrightarrow B_n, \\ T_{a, q}(v_{\rho, r} t^i) &= (-1)^{\lfloor qi/n \rfloor} a^\rho \sum_{\ell=0}^{m_\rho-1-r} \frac{(\log a)^\ell}{\ell!} v_{\rho, r+\ell} t^{qi \bmod n}. \end{aligned} \quad (\text{MW9a})$$

Here A_a and N act coefficientwise, and B_n is the direct sum of the $B_{\rho, n}$. Applying MW8 to MW9 gives every coefficient in MW9a; the two orders agree because the signed permutation acts only on the coefficient-slot index. The split lift of $T_{a, q}$ sends the face labelled A to the face labelled qA , with exactly the displayed signs. Eigenspaces below are taken in the full linear space $B_{\rho, n}$; an individual moving face need not be invariant.

Since q is a unit modulo $2n$, choose $h \geq 1$ with $q^h \equiv 1 \pmod{2n}$. Substitution sends t^i to $t^{q^h i} = t^i$, so $\Psi_q^h = I$. The polynomial $X^h - 1$ has distinct complex roots, hence Ψ_q is diagonalizable and its eigenvalues ω satisfy $\omega^h = 1$. On each nonzero graded part of its ω -eigenspace, the three separate operators therefore have eigenvalues

$$\text{Gr}^M(A_a) : a^\rho, \quad \text{Gr}^M(\Psi_q) : \omega, \quad \text{Gr}^M(T_{a, q}) : \omega a^\rho, \quad |\omega| = 1. \quad (\text{MW10})$$

The eigenspace decomposition respects the filtration. Its projection onto the ω -eigenspace is

$$P_\omega = h^{-1} \sum_{d=0}^{h-1} \omega^{-d} \Psi_q^d.$$

On an eigenspace with eigenvalue ω' , this sum is one if $\omega' = \omega$ and zero otherwise, by the finite geometric-sum identity and $(\omega'/\omega)^h = 1$. Every summand preserves every filtration step, so the projection does too. Thus the asserted graded eigenspaces and the restrictions of A_a are defined. MW10 computes the combined operator's modulus as $a^{\text{Re } \rho}$; no bound on $\text{Re } \rho$ has been inserted.

1.3 Tensor products with every coefficient and sign retained

The following proof supplies the tensor compatibility on these blocks, including its constants. On a block of length m , define auxiliary operators

$$Hv_r = (m - 1 - 2r)v_r, \quad Lv_r = v_{r+1}, \quad Ev_r = r(m - r)v_{r-1}. \quad (\text{MW11})$$

The formulas at the endpoints use $v_{-1} = v_m = 0$; L is precisely the original nilpotent N_ρ . Direct subtraction on each basis vector gives

$$[H, L] = -2L, \quad [H, E] = 2E, \quad [E, L] = H. \quad (\text{MW12})$$

For example the coefficient in the last commutator is $(r+1)(m-r-1) - r(m-r) = m-1-2r$. For this algebraic proof alone equip this basis with the auxiliary positive Hermitian form having squared lengths

$$c_r = \frac{r!(m-1)!}{(m-1-r)!}, \quad c_0 = 1. \quad (\text{MW13})$$

This is an explicitly specified additional form, not a replacement for a source norm or an arithmetic positivity assertion. The identity $c_{r+1} = (r+1)(m-r-1)c_r$ proves $L^* = E$, and $H^* = H$.

On any finite tensor product of blocks, retain the original tensor basis and define $L_{\text{tot}}, E_{\text{tot}}, H_{\text{tot}}$ as the sums of these operators acting in the individual factors. Operators on different factors commute, so MW12 still holds. With the product form from MW13, $L_{\text{tot}}^* = E_{\text{tot}}$ and $H_{\text{tot}}^* = H_{\text{tot}}$.

Here is a direct proof of the needed graded isomorphisms. Choose a nonzero vector v of maximal H_{tot} -eigenvalue d . Then $E_{\text{tot}}v = 0$, because E_{tot} raises that eigenvalue by two. Induction using MW12 gives

$$E_{\text{tot}}L_{\text{tot}}^r v = r(d-r+1)L_{\text{tot}}^{r-1}v. \quad (\text{MW14})$$

The operator L_{tot} is nilpotent. Let ℓ be the largest exponent for which $L_{\text{tot}}^\ell v \neq 0$. Applying MW14 at $r = \ell + 1$ gives $(\ell+1)(d-\ell)L_{\text{tot}}^\ell v = 0$, hence $d = \ell$. Thus d is a nonnegative integer and the chain has weights $d, d-2, \dots, -d$. Its span is invariant under $L_{\text{tot}}, E_{\text{tot}}, H_{\text{tot}}$. Its orthogonal complement is invariant too, because the first two are adjoints and the third is self-adjoint. Induction on dimension decomposes the entire tensor space into such chains. On each chain L_{tot}^k identifies weight k with weight $-k$ exactly as in MW4. Therefore the filtration by sums of H_{tot} -eigenspaces of index at most j is the centered nilpotent filtration. In the original tensor coordinates it is

$$M_j \left(\bigotimes_{\nu=1}^k E_{\rho_\nu} \right) = \sum_{j_1 + \dots + j_k \leq j} M_{j_1} E_{\rho_1} \otimes \dots \otimes M_{j_k} E_{\rho_k}. \quad (\text{MW15})$$

This also proves the precise relation to Weil II 1.6.9 on these actual blocks. Tensoring labelled faces retains the ordered tuple of their masks; MW15 operates on their amplitude fibres.

The original arithmetic tensor operator is, without dropping any term,

$$\bigotimes_{\nu=1}^k A_a|_{E_{\rho_\nu}} = a^{\rho_1 + \dots + \rho_k} \exp \left((\log a) \sum_{\nu=1}^k N_{\rho_\nu} \right). \quad (\text{MW16})$$

The exponential is finite and the equality follows by expanding the product of commuting finite exponentials. On the M -graded pieces its eigenvalue is $a^{\rho_1 + \dots + \rho_k}$. For identical packets its modulus is $a^{k \text{Re } \rho}$. In particular the original arithmetic weight survives this complete nilpotent-filtration construction.

For the coefficient spaces, take $n_\nu \geq 1$, a common odd prime power q coprime to every n_ν , and $B_\nu = E_{\rho_\nu}[t_\nu]/(t_\nu^{n_\nu} + 1)$. Write N_ν for the nilpotent in factor ν , extended by identities to the other factors. Define $A_a^\otimes = \bigotimes_\nu A_a|_{B_\nu}$, $\Psi_q^\otimes = \bigotimes_\nu \Psi_q|_{B_\nu}$, and $T_{a,q}^\otimes = \bigotimes_\nu T_{a,q}|_{B_\nu}$. Then

$$\begin{aligned} T_{a,q}^\otimes &= A_a^\otimes \Psi_q^\otimes = a^{\sum_\nu \rho_\nu} \exp\left((\log a) \sum_\nu N_\nu\right) \Psi_q^\otimes, \\ T_{a,q}^\otimes \left(\bigotimes_\nu v_{\rho_\nu, r_\nu} t_\nu^{i_\nu}\right) &= a^{\sum_\nu \rho_\nu} \left(\prod_\nu (-1)^{\lfloor q i_\nu / n_\nu \rfloor}\right) \\ &\quad \times \sum_{0 \leq \ell_\nu \leq m_{\rho_\nu} - 1 - r_\nu \text{ for all } \nu} \left(\prod_\nu \frac{(\log a)^{\ell_\nu}}{\ell_\nu!}\right) \bigotimes_\nu v_{\rho_\nu, r_\nu + \ell_\nu} t_\nu^{q i_\nu \bmod n_\nu}. \end{aligned} \tag{MW16a}$$

Expanding the factor formulas MW9a proves the second line, including every sign and every factorial. The first line follows by grouping the commuting operators on different tensor factors, and then using MW16. On the tensor of coefficient eigenspaces with eigenvalues ω_ν , the three graded eigenvalues are respectively

$$A_a^\otimes : a^{\sum_\nu \rho_\nu}, \quad \Psi_q^\otimes : \prod_\nu \omega_\nu, \quad T_{a,q}^\otimes : \left(\prod_\nu \omega_\nu\right) a^{\sum_\nu \rho_\nu}. \tag{MW16b}$$

The positive nilpotent powers in MW16a decrease total degree by at least two, which proves this graded formula from the complete operator formula. Its combined modulus is $a^{\sum_\nu \text{Re } \rho_\nu}$. The coefficient permutation sends the ordered tuple $(A_\nu)_\nu$ to $(qA_\nu)_\nu$.

1.4 The exact Tate-scaling comparison

Deligne's local-monodromy operator has type $N : V(1) \rightarrow V$. For geometric Frobenius over $\mathbb{F}_q$, the Tate line has Frobenius eigenvalue q^{-1} . After recording a generator of that line, equivariance has the precise matrix form

$$FNF^{-1} = q^{-1}N. \tag{MW17}$$

For the original arithmetic action MW8 one instead calculates $A_a N_\rho A_a^{-1} = N_\rho$. The exact relation between these operator presentations can be constructed on the same jet space. For real $q > 1$ define

$$\Gamma_{q,\rho} v_{\rho,r} = q^{(m_\rho-1)/2-r} v_{\rho,r}, \quad F_{q,\rho} = q^\rho \Gamma_{q,\rho}. \tag{MW18}$$

Both are invertible and retain every original basis vector. The quotient of successive diagonal entries is q^{-1} , so

$$F_{q,\rho} N_\rho F_{q,\rho}^{-1} = q^{-1} N_\rho, \quad A_q = F_{q,\rho} \Gamma_{q,\rho}^{-1} \exp((\log q) N_\rho). \tag{MW19}$$

The second equality is obtained by multiplying the first two diagonal operators, giving exactly $q^\rho I$, and then retaining the complete exponential MW8. It proves reconstruction of the original arithmetic action from the added operators. Moreover, if $U(z) = \exp(z N_\rho)$, then

$$F_{q,\rho} U(z) F_{q,\rho}^{-1} = U(q^{-1} z). \tag{MW20}$$

Thus these explicit matrices define a complex representation of the group generated by the additive one-parameter operators $U(z)$ and an invertible element acting on the parameter by $z \mapsto q^{-1} z$. This statement is at the displayed complex-linear type; it does not assert that these matrices arise from an arithmetic lisse sheaf.

On the graded piece of index $j = m_\rho - 1 - 2r$, the eigenvalue of the added $F_{q,\rho}$ is $q^{\rho+j/2}$, with

$$2 \log_q |q^{\rho+j/2}| = 2 \text{Re } \rho + j. \tag{MW21}$$

This gives exactly the index shift $\beta + j$ in Weil II 1.8.4, with the explicitly computed $\beta = 2 \text{Re } \rho$. It does not determine β . The two-sided boundary estimate in Deligne's theorem comes from the

pure lisse source, the trace and Euler-product bound, and its tensor and dual arguments. MW18 constructs the matrix relation and MW21 computes every graded modulus; neither computation supplies that geometric input. All operators MW18 act coefficientwise on MW7, preserve its masks, and commute with the coefficient permutation MW9.

For admissible odd prime powers q as in MW9, retaining the separate coefficient operator also gives the exact reconstruction and its tensor version

$$\begin{aligned} T_{q,q}|_{B_{\rho,n}} &= F_{q,\rho} \Gamma_{q,\rho}^{-1} \exp((\log q) N_\rho) \Psi_q, \\ T_{q,q}^\otimes &= F_q^\otimes (\Gamma_q^\otimes)^{-1} \exp\left((\log q) \sum_\nu N_\nu\right) \Psi_q^\otimes, \\ F_q^\otimes &= \bigotimes_\nu F_{q,\rho_\nu}, \quad \Gamma_q^\otimes = \bigotimes_\nu \Gamma_{q,\rho_\nu}. \end{aligned} \tag{MW21a}$$

The first two factors in the first line multiply to $q^\rho I$; in the tensor line they multiply to $q^{\sum_\nu \rho_\nu} I$. This proves both identities in their displayed order. On total degree j , the added $F_q^\otimes$ acts by $q^{\sum_\nu \rho_\nu + j/2}$. Its combination with $\Psi_q^\otimes$ acts by $(\prod_\nu \omega_\nu) q^{\sum_\nu \rho_\nu + j/2}$ on the indicated coefficient eigenspaces. These are added-operator eigenvalues; the unchanged arithmetic and combined eigenvalues remain MW16b.

For completeness, duality at the vector-space level also retains its signs. On $V^\vee$, use $N^\vee = -N^\top$ and $M_j V^\vee = (M_{-j-1} V)^\perp$. The dual of a basis vector of index d has index $-d$, and $N^\vee$ lowers that index by two with coefficient -1 . Its iterated graded isomorphisms have the nonzero coefficient $(-1)^k$. The original contragredient arithmetic action is

$$(A_a^{-1})^\top = a^{-\rho} \exp(-(\log a) N^\top), \tag{MW22}$$

so its graded eigenvalue is exactly $a^{-\rho}$. On the full coefficient dual put $\Psi_q^\vee = (\Psi_q^{-1})^\top$ and $T_{a,q}^\vee = (T_{a,q}^{-1})^\top$. Commutativity in MW9a and transposition of the inverse product give

$$T_{a,q}^\vee = (A_a^{-1})^\top (\Psi_q^{-1})^\top. \tag{MW22a}$$

The dual of the ω -coefficient eigenspace has $\Psi_q^\vee$ -eigenvalue ω^{-1} : evaluation gives $(\Psi_q^\vee f)(v) = f(\Psi_q^{-1} v) = \omega^{-1} f(v)$. Thus the three dual graded eigenvalues are $a^{-\rho}$, ω^{-1} , and $\omega^{-1} a^{-\rho}$, respectively. These formulas establish the tensor, dual and Tate-scaling comparisons on every arithmetic amplitude fibre while retaining the separate mixed-support labels.

1.5 The actual length-two quotient retains its induced degrees

Define $Z_{\geq 2} = \{\rho \in Z : m_\rho \geq 2\}$. For each member of this specified subset put $m = m_\rho$, $b = m - 2$, and

$$\begin{aligned} J_\rho &= u_\rho^2 E_\rho = \text{span}\{v_{\rho,r} : 2 \leq r < m\}, \quad C_\rho = E_\rho / J_\rho, \quad \pi_\rho : E_\rho \longrightarrow C_\rho, \\ w_0 &= \pi_\rho(v_{\rho,0}), \quad w_1 = \pi_\rho(v_{\rho,1}), \quad \phi_\rho : C_\rho \longrightarrow V_2 = \mathbb{C}[u]/(u^2), \quad \phi_\rho(c_0 w_0 + c_1 w_1) = c_0 + c_1 u. \end{aligned} \tag{MW23}$$

Here $u = x - \rho$ is the same local coordinate: the map sends the original local unit to 1 and the original first power to u . The basis multiplication satisfies $v_{\rho,r} v_{\rho,s} = v_{\rho,r+s}$ for $r + s < m$, and zero otherwise. Therefore J_ρ is an ideal, w_0 is the quotient unit, $w_1^2 = 0$, and ϕ_ρ is an algebra isomorphism with inverse $c_0 + c_1 u \mapsto c_0 w_0 + c_1 w_1$. For $m = 2$, the displayed span for J_ρ is zero. This construction is indexed by $Z_{\geq 2}$; it does not assert that this subset has a member. Raising an exponent preserves J_ρ , so N_ρ induces the operator $\bar{N}$ on C_ρ .

Give J_ρ the subspace filtration and C_ρ the quotient filtration of MW3. The latter is exactly

$$\begin{aligned} \bar{M}_j C_\rho &= \pi_\rho(M_j E_\rho) = \begin{cases} 0 & j < m - 3, \\ \mathbb{C} w_1 & m - 3 \leq j < m - 1, \\ C_\rho & j \geq m - 1, \end{cases} \\ \phi_\rho(\bar{M}_j C_\rho) &= M_{j-b} V_2. \end{aligned} \tag{MW24}$$

Indeed, all original powers of exponent at least two vanish under π_ρ ; the two surviving powers have their original degrees $m-1$ and $m-3$. Substituting $b = m-2$ into the length-two degrees $1, -1$ proves the second line. For every j the kernel of $M_j E_\rho \rightarrow \overline{M}_j C_\rho$ is $J_\rho \cap M_j E_\rho$, and its image is the whole indicated quotient step. Hence $0 \rightarrow J_\rho \cap M_j E_\rho \rightarrow M_j E_\rho \rightarrow \overline{M}_j C_\rho \rightarrow 0$ is exact, proving strictness of both maps. The quotient filtration is centered at b : its only nonidentity graded isomorphism is $\overline{N} : \text{Gr}_{b+1}^{\overline{M}} C_\rho \rightarrow \text{Gr}_{b-1}^{\overline{M}} C_\rho$, sending the class of w_0 to that of w_1 with coefficient one. For $k=0$ the centered-at- b map is the identity, and for $k>1$ both graded spaces are zero. If $m>2$, this filtration is not centered at zero: Gr_{m-1} is nonzero whereas $\text{Gr}_{-(m-1)}$ is zero. MW24 retains the quotient's actual degrees in every case.

The ideal J_ρ is invariant under N_ρ , every A_a , and the added diagonal operators $\Gamma_{q,\rho}, F_{q,\rho}$. For N_ρ this follows by raising the exponent; for A_a it follows term by term in MW8; for the diagonals each basis line is invariant. Their induced quotient operators consequently satisfy

$$\begin{aligned} \overline{N}w_0 &= w_1, \quad \overline{N}w_1 = 0, \quad \overline{A}_a w_0 = a^\rho(w_0 + (\log a)w_1), \quad \overline{A}_a w_1 = a^\rho w_1, \\ \overline{\Gamma}_q w_r &= q^{(m-1)/2-r} w_r, \quad \overline{F}_q w_r = q^{\rho+(m-1)/2-r} w_r \quad (r=0,1), \\ \phi_\rho \overline{N} &= N_2 \phi_\rho, \quad \phi_\rho \overline{A}_a = A_{a,2} \phi_\rho, \quad \phi_\rho \overline{\Gamma}_q = q^{b/2} \Gamma_{q,2} \phi_\rho, \quad \phi_\rho \overline{F}_q = q^{b/2} F_{q,2} \phi_\rho. \end{aligned} \tag{MW25}$$

The subscript 2 specifies the fixed length-two operators $N_2(1) = u$, $N_2(u) = 0$, $A_{a,2} = a^\rho(I + (\log a)N_2)$, $\Gamma_{q,2}(1) = q^{1/2}$, $\Gamma_{q,2}(u) = q^{-1/2}u$, and $F_{q,2} = q^\rho \Gamma_{q,2}$. Every line of MW25 follows by applying both sides to w_0, w_1 . In particular, the higher terms of MW8 are killed by this specified quotient map; they have not been discarded from the original packet.

An exact filtered operator isomorphism records both the degree and diagonal scalar shifts. Let $D_b = \mathbb{C}d_b$ have filtration $(D_b)_j = 0$ for $j < b$ and $(D_b)_j = D_b$ for $j \geq b$, with $N_{D_b} = 0$, $A_{a,D_b} = I$, and $\Gamma_{q,D_b} = F_{q,D_b} = q^{b/2}I$. Use sum filtrations, tensor-sum nilpotents, and tensor-product invertible operators. Then

$$\begin{aligned} \theta_\rho : C_\rho &\xrightarrow{\sim} V_2 \otimes D_b, \quad w_r \mapsto u^r \otimes d_b \quad (r=0,1), \\ \theta_\rho(\overline{M}_j C_\rho) &= M_{j-b} V_2 \otimes D_b, \quad \overline{A}_q = \overline{F}_q \overline{\Gamma}_q^{-1} (I + (\log q) \overline{N}). \end{aligned} \tag{MW26}$$

The inverse sends $u^r \otimes d_b$ to w_r , so the displayed filtration equality proves strictness in both directions. MW25 proves intertwining for A, Γ, F , and its coefficient-one nilpotent relation proves intertwining for N . In the reconstruction, $\overline{F}_q \overline{\Gamma}_q^{-1} = q^\rho I$, so the degree shift has changed no arithmetic scalar or logarithmic coefficient. The original auxiliary H in MW11 descends to $H_2 + bI$ under ϕ_ρ , where $H_2(1) = 1$ and $H_2(u) = -u$, by its two diagonal entries. For $m>2$, the original auxiliary E does not descend: $E(v_{\rho,2}) = 2(m-2)v_{\rho,1} \notin J_\rho$. Thus MW26 is the explicitly proved filtered isomorphism for N, A, Γ, F , with the exact additional relation for H .

The kernel also has a specified shifted comparison. For $m>2$, let $V_{m-2} = \mathbb{C}[u]/(u^{m-2})$ use the original powers, the centered filtration of a length- $m-2$ block, and the operators MW8 and MW18 with the same ρ . The linear map $\alpha_\rho : V_{m-2} \rightarrow J_\rho$, $\alpha_\rho(u^r) = v_{\rho,r+2}$, is bijective, with inverse given by removing these specified two powers. It is a module comparison for the displayed nilpotent and arithmetic operators. Its complete filtration and diagonal relations are

$$\begin{aligned} \alpha_\rho(M_{j+2}V_{m-2}) &= J_\rho \cap M_j E_\rho, \quad N_\rho \alpha_\rho = \alpha_\rho N_{m-2}, \quad A_a \alpha_\rho = \alpha_\rho A_{a,m-2}, \\ \Gamma_{q,\rho} \alpha_\rho &= q^{-1} \alpha_\rho \Gamma_{q,m-2}, \quad F_{q,\rho} \alpha_\rho = q^{-1} \alpha_\rho F_{q,m-2}. \end{aligned} \tag{MW26a}$$

The source degree of u^r is $m-3-2r$; the image's degree is $m-1-2(r+2) = (m-3-2r)-2$, proving the first relation at every step. Raising the exponent proves the nilpotent relation including the last zero vector; expanding MW8 proves the arithmetic relation with all its coefficients. The diagonal exponent on the image is $(m-1)/2 - (r+2) = ((m-3)/2 - r) - 1$, which proves both remaining relations. Consequently the kernel, quotient, and their exact sequence all retain their actual induced shifts. For $m=2$ the kernel is zero as already specified.

The same quotient is available coefficientwise. Its map on each face is

$$(A, (x_i)_{i \in A}) \mapsto (A, (\pi_\rho x_i)_{i \in A}).$$

Even a coefficient in J_ρ becomes a present zero in the same slot. Surjectivity on each filtered face follows by choosing, independently in each slot, a preimage in $M_j E_\rho$ from MW24. Zero insertion commutes with this map because $\pi_\rho(0) = 0$. On the full coefficient space it intertwines Ψ_q since the map is linear and sends $(-1)^{\lfloor qi/n \rfloor} x$ to that same sign times $\pi_\rho(x)$; MW25 then proves intertwining for $T_{a,q}$. For a tensor of such quotient blocks the exact comparison is

$$\bigotimes_{\nu} C_{\rho_{\nu}} \xrightarrow{\bigotimes_{\nu} \theta_{\rho_{\nu}}} \left(\bigotimes_{\nu} V_{2, \rho_{\nu}} \right) \otimes \left(\bigotimes_{\nu} D_{m_{\rho_{\nu}} - 2} \right), \quad (\text{MW27})$$

total degree shift $\sum_{\nu} (m_{\rho_{\nu}} - 2)$, added F and Γ factor $q^{\frac{1}{2} \sum_{\nu} (m_{\rho_{\nu}} - 2)}$.

The tensor factors are rearranged by the explicit ordinary vector-space map $\bigotimes_{\nu} (u^{r_{\nu}} \otimes d_{b_{\nu}}) \mapsto (\bigotimes_{\nu} u^{r_{\nu}}) \otimes (\bigotimes_{\nu} d_{b_{\nu}})$, with coefficient one. Degrees add and diagonal eigenvalues multiply, which proves both formulas and their inverse on every tensor basis vector. The tensor quotient is strict: a source tensor basis vector survives exactly when each exponent is zero or one, and the degree of every surviving vector is unchanged. Thus each filtered target step is the image of the same source step. Its kernel is spanned by tensors with at least one exponent at least two, giving the complete filtered kernel as their intersection with that step. The unchanged arithmetic eigenvalues and the signed coefficient products are precisely MW16a–MW16b, with the nilpotent powers transported through these actual quotient maps. The strict length-two subobject and quotient maps are developed with these induced degrees in MRE34–MRE42 below.

Status of the geometric control. The construction proves MW1–MW27, including the lettered formulas, on the original finite packets and the specified quotients. It explicitly supplies the nilpotent filtration and its tensor and dual compatibility, and it reconstructs the unchanged arithmetic operator from a Tate-scaling operator. No theorem here identifies the resulting complex representation with a pure lisse sheaf over the absolute τ -base. In particular it proves no numerical restriction on $2 \operatorname{Re} \rho$. The relevant further comparison is with the actual hypotheses and proof of Weil II 1.8.4–1.8.5 and the compact-support pushforward estimate 3.3.1; the finite-poset section functor by itself has not supplied those estimates.

2 Strict filtered control of the original length-two jet extension

This calculation constructs the induced filtrations and exact operator maps on a specified arithmetic jet extension. It includes the uniqueness proof for the centered nilpotent filtration, the translation to a specified constituent degree, and the relative filtration on this extension. The original arithmetic action and the added Tate-scaling action are both retained as separate, explicitly related operators.

2.1 Uniqueness of the centered nilpotent filtration

Let V be a finite-dimensional complex vector space and N a nilpotent endomorphism. A finite increasing filtration in this paragraph means subspaces $M_j V$, indexed by $j \in \mathbb{Z}$, which are zero for all sufficiently negative j and equal to V for all sufficiently positive j . Suppose it has the centered nilpotent properties

$$N M_j V \subseteq M_{j-2} V, \quad N^k : \operatorname{Gr}_k^M V \xrightarrow{\sim} \operatorname{Gr}_{-k}^M V \quad (k \geq 0). \quad (\text{MRE1})$$

Here and below an isomorphism from the zero vector space to itself is included in the assertion.

We prove that these properties determine M uniquely. Choose $d \geq 0$ with $N^{d+1} = 0$. For $k > d$, the isomorphism in MRE1 is the zero map; its source and target therefore vanish. Exhaustiveness and finiteness give $M_d V = V$ and $M_{-d-1} V = 0$. The isomorphism for $k = d$ is induced by N^d from $V/M_{d-1} V$ to $M_{-d} V$. Consequently

$$M_{d-1} V = \ker N^d, \quad M_{-d} V = \operatorname{im} N^d. \quad (\text{MRE2})$$

For $d = 0$, this says that the only possibly nonzero graded piece is degree zero, and determines the filtration completely.

For $d \geq 1$, the subspace $\text{im } N^d$ lies in $\ker N^d$, because $N^{2d} = 0$. The induced operator on

$$K_d = \ker N^d / \text{im } N^d \quad (\text{MRE3})$$

satisfies $N^d = 0$. The filtration induced on K_d has the same graded pieces as V for $-d < j < d$, and zero graded pieces outside this range. Thus MRE1 descends to this quotient: for $0 \leq k < d$ its maps are the original graded maps, and for $k \geq d$ both sides vanish. Induction on d determines the filtration on K_d . Every intermediate step on V is its inverse image under the quotient $\ker N^d \rightarrow K_d$; MRE2 fixes the endpoints. This proves uniqueness.

For the original length- m jet basis v_r , $0 \leq r < m$, with $Nv_r = v_{r+1}$ and $Nv_{m-1} = 0$, existence is explicit:

$$M_j V = \text{span}\{v_r : m - 1 - 2r \leq j\}. \quad (\text{MRE4})$$

The operator lowers the displayed index by two. When index $k \geq 0$ occurs, its basis vector has exponent $r = (m - 1 - k)/2$; applying N^k gives the index- $-k$ vector v_{r+k} , with coefficient one and $r + k = m - 1 - r$. If index k is absent, index $-k$ is absent as well. This proves MRE1. Taking direct sums proves existence on any finite packet of these original blocks. The preceding uniqueness proof identifies this explicit filtration with the centered nilpotent filtration.

2.2 Translation to a specified constituent degree

For an integer b , put a given original block in the one-step filtration

$$W_i^{[b]} V = \begin{cases} 0 & i < b, \\ V & i \geq b, \end{cases} \quad M_j^{[b]} V = M_{j-b} V. \quad (\text{MRE5})$$

The relative nilpotent-filtration identities for this specified $W^{[b]}$ are

$$NM_j^{[b]} \subseteq M_{j-2}^{[b]}, \quad N^k : \text{Gr}_{i+k}^{M^{[b]}} \text{Gr}_i^{W^{[b]}} V \xrightarrow{\sim} \text{Gr}_{i-k}^{M^{[b]}} \text{Gr}_i^{W^{[b]}} V. \quad (\text{MRE6})$$

For $i \neq b$, both sides of the second map vanish. For $i = b$, that map is exactly the already proved $N^k : \text{Gr}_k^M V \rightarrow \text{Gr}_{-k}^M V$. The first inclusion follows directly by shifting both indices in MRE1. This proves the relative-filtration translation for every integer b . The construction of these filtrations assigns the displayed degree; it does not change or determine any arithmetic eigenvalue.

2.3 The original extension and its arithmetic action

Fix $\rho \in \mathbb{C}$, retain the local coordinate $u = x - \rho$, and define the length-two jet algebra and its nilpotent operator by

$$V_2 = \mathbb{C}[u]/(u^2), \quad v_0 = 1, \quad v_1 = u, \quad Nv_0 = v_1, \quad Nv_1 = 0. \quad (\text{MRE7})$$

In this entire calculation these are the fixed original basis vectors and the coefficient of Nv_0 remains exactly one. For arbitrary ρ , MRE7 specifies this displayed local algebra; it asserts no zero or repeated zero of $g(s) = 2\xi(s)$. The exact comparison with any actual packet member of multiplicity at least two is proved in MRE34–MRE42. That comparison retains the quotient filtration induced from the full block, together with its degree translation and both added diagonal factors. Let $K = \mathbb{C}u \subset V_2$, let $Q = V_2/K$, and write $\bar{1}$ for the class of 1 in Q . The maps

$$i : K \rightarrow V_2, \quad i(cu) = cu, \quad p : V_2 \rightarrow Q, \quad p(a + bu) = a\bar{1} \quad (\text{MRE8})$$

give the exact sequence

$$0 \rightarrow K \xrightarrow{i} V_2 \xrightarrow{p} Q \rightarrow 0. \quad (\text{MRE9})$$

Indeed, i is injective, p is surjective, and $\ker p = \mathbb{C}u = \operatorname{im} i$. The restricted operator on K and the induced operator on Q are both zero.

For real $a > 0$, retain the original arithmetic action

$$A_{a,2} = a^\rho(I + (\log a)N), \quad A_{a,K} = a^\rho I_K, \quad A_{a,Q} = a^\rho I_Q, \quad a^\rho = \exp(\rho \log a). \quad (\text{MRE10})$$

Since $N^2 = 0$, this is exactly the complete original finite exponential. It gives

$$A_{a,2}(c + du) = a^\rho(c + (d + c \log a)u). \quad (\text{MRE11})$$

The restriction to K and the induced quotient action are therefore the displayed scalars. In particular

$$A_{a,2}i = iA_{a,K}, \quad pA_{a,2} = A_{a,Q}p. \quad (\text{MRE12})$$

All maps in MRE9 preserve these original arithmetic actions.

2.4 Induced subobject and quotient filtrations

The centered filtration MRE4 on V_2 is

$$M_j V_2 = \begin{cases} 0 & j < -1, \\ K & -1 \leq j < 1, \\ V_2 & j \geq 1. \end{cases} \quad (\text{MRE13})$$

Give the subobject and quotient their actual induced filtrations:

$$M_j^K = K \cap M_j V_2 = \begin{cases} 0 & j < -1, \\ K & j \geq -1, \end{cases} \quad M_j^Q = p(M_j V_2) = \begin{cases} 0 & j < 1, \\ Q & j \geq 1. \end{cases} \quad (\text{MRE14})$$

Let $C_j K$ and $C_j Q$ denote the centered filtrations of the zero nilpotent on these one-dimensional spaces: each is zero for $j < 0$ and the whole space for $j \geq 0$. Then the precise shifts are

$$M_j^K = C_{j+1} K, \quad M_j^Q = C_{j-1} Q. \quad (\text{MRE15})$$

Thus the subobject retains degree -1 , the quotient retains degree $+1$, and the middle object's filtration remains MRE13.

For every integer j , the sequence

$$0 \longrightarrow M_j^K \xrightarrow{i} M_j V_2 \xrightarrow{p} M_j^Q \longrightarrow 0 \quad (\text{MRE16})$$

is exact. For $j < -1$, all three spaces are zero. For $-1 \leq j < 1$, it is $0 \rightarrow K \xrightarrow{1} K \rightarrow 0 \rightarrow 0$. For $j \geq 1$, it is MRE9. This exhaustive calculation also proves strictness: the image of the filtered inclusion is exactly $\operatorname{im} i \cap M_j V_2$, and $p(M_j V_2) = M_j^Q$. On associated graded, the inclusion identifies the two degree -1 spaces, the quotient identifies the two degree $+1$ spaces, and all other graded spaces vanish.

The relation to the separately centered one-dimensional filtrations is also explicit. Their step -1 is zero, whereas $K \cap M_{-1} V_2 = K$. Therefore the original inclusion with the separately centered filtration is filtration preserving but is not strict. Formula MRE15 supplies the exact shifted subobject and quotient filtrations which restore the strict filtered exact sequence MRE16.

2.5 Added Frobenius and the exact shifted intertwiners

For real $q > 1$, use the additional diagonal operators

$$\Gamma_{q,2} v_0 = q^{1/2} v_0, \quad \Gamma_{q,2} v_1 = q^{-1/2} v_1, \quad F_{q,2} = q^\rho \Gamma_{q,2}. \quad (\text{MRE17})$$

Their induced operators on MRE9 are

$$\begin{aligned}\Gamma_{q,K} &= q^{-1/2}I_K, & F_{q,K} &= q^{\rho-1/2}I_K, \\ \Gamma_{q,Q} &= q^{1/2}I_Q, & F_{q,Q} &= q^{\rho+1/2}I_Q.\end{aligned}\tag{MRE18}$$

Every displayed operator preserves the indicated filtration. Directly on u and $\bar{1}$, the inclusion and quotient satisfy

$$F_{q,2}i = iF_{q,K}, \quad pF_{q,2} = F_{q,Q}p, \quad \Gamma_{q,2}i = i\Gamma_{q,K}, \quad p\Gamma_{q,2} = \Gamma_{q,Q}p.\tag{MRE19}$$

This proves exact filtered intertwining with the unchanged extension maps. If $F_{q,K}^{\text{cen}} = q^\rho I_K$ and $F_{q,Q}^{\text{cen}} = q^\rho I_Q$ are the operators obtained by assigning the one-dimensional spaces their separate centered degree zero, the same equations read

$$F_{q,2}i = q^{-1/2}iF_{q,K}^{\text{cen}}, \quad pF_{q,2} = q^{1/2}F_{q,Q}^{\text{cen}}p.\tag{MRE20}$$

The degree shifts and the scalar factors are therefore exactly the same correction, in filtration and operator form respectively.

The original arithmetic operators are reconstructed without replacing their eigenvalues:

$$\begin{aligned}A_{q,2} &= F_{q,2}\Gamma_{q,2}^{-1}(I + (\log q)N), \\ A_{q,K} &= F_{q,K}\Gamma_{q,K}^{-1} = q^\rho I_K, \\ A_{q,Q} &= F_{q,Q}\Gamma_{q,Q}^{-1} = q^\rho I_Q.\end{aligned}\tag{MRE21}$$

The first line has the displayed multiplication order. In all three lines $F\Gamma^{-1}$ is exactly q^ρ times the identity. Thus the shifted added eigenvalues retain, through these exact formulas, the original arithmetic scalar and its complete nilpotent term.

The added graded moduli are fully determined:

$$2\log_q |F_{q,K}| = 2\operatorname{Re} \rho - 1, \quad 2\log_q |F_{q,Q}| = 2\operatorname{Re} \rho + 1.\tag{MRE22}$$

Here $|F|$ means the modulus of the displayed one-dimensional eigenvalue. Equation MRE22 records the exact shift; it imposes no restriction on $\operatorname{Re} \rho$.

2.6 The nilpotent link and its Tate-typed isomorphism

Since N vanishes on K and maps V_2 into K , it factors uniquely through the quotient as

$$N = i\bar{N}p, \quad \bar{N} : Q \xrightarrow{\sim} K, \quad \bar{N}(\bar{1}) = u.\tag{MRE23}$$

The factorization follows by applying both sides to $a + bu$, where both sides give au . The factor map is bijective because it sends the specified basis vector to the specified basis vector with coefficient one. It lowers the induced filtration degree by exactly two. It commutes with the original arithmetic scalars and satisfies the exact added-operator relation

$$A_{a,K}\bar{N} = \bar{N}A_{a,Q}, \quad F_{q,K}\bar{N} = q^{-1}\bar{N}F_{q,Q}.\tag{MRE24}$$

The second equality follows from $q^{\rho-1/2} = q^{-1}q^{\rho+1/2}$.

Define the explicit one-dimensional complex character $T_q = \mathbb{C}t$ with its chosen generator t , filtration concentrated in degree -2 , and operators

$$F_{q,T}t = q^{-1}t, \quad \Gamma_{q,T}t = q^{-1}t, \quad A_{a,T}t = t \quad (a > 0).\tag{MRE25}$$

The added eigenvalue is the geometric-Frobenius Tate character. At this stage T_q is the displayed complex filtered line; no arithmetic sheaf realization is required for the following proved matrix map. Give tensor products the sum filtration and tensor product operators. Then $Q \otimes T_q$ is concentrated

in degree $1 - 2 = -1$, with added eigenvalue $q^{\rho+1/2}q^{-1} = q^{\rho-1/2}$ and original arithmetic eigenvalue a^ρ . Therefore

$$\nu : Q \otimes T_q \xrightarrow{\sim} K, \quad \nu(\bar{1} \otimes t) = u \quad (\text{MRE26})$$

is a strict filtered isomorphism intertwining F_q , Γ_q , and every A_a . Its inverse sends u to $\bar{1} \otimes t$, so these claims include both inverse maps. Formula MRE26 is the exact Tate-typed form of the original nilpotent link MRE23.

The map N itself has the same strict filtered type. On $V_2 \otimes T_q$, define $\mathcal{N}(v \otimes t) = Nv$. The domain filtration is $M_j(V_2 \otimes T_q) = M_{j+2}V_2 \otimes T_q$. For every j ,

$$N(M_{j+2}V_2) = \text{im } N \cap M_jV_2. \quad (\text{MRE27})$$

For $j < -1$, both sides vanish: the domain step is at most K , which N kills. For $j \geq -1$, the domain step is V_2 , its image is K , and the right side is also K . This proves strictness. The relation $F_{q,2}NF_{q,2}^{-1} = q^{-1}N$, checked on v_0, v_1 , proves F_q -equivariance of $\mathcal{N}$. The corresponding relation for Γ_q proves its equivariance, and $A_{a,2}N = NA_{a,2}$ proves equivariance for every original arithmetic action.

2.7 The explicit relative filtration on this extension

Retain the extension filtration with its two induced constituent degrees:

$$W_jV_2 = \begin{cases} 0 & j < -1, \\ K & -1 \leq j < 1, \\ V_2 & j \geq 1. \end{cases} \quad (\text{MRE28})$$

This is a specified filtered extension, with degree- -1 constituent K and degree- $+1$ constituent Q . Set $R_jV_2 = M_jV_2$, with M as in MRE13. Then $R = W$ as displayed filtrations, and R is the relative nilpotent filtration for this specified W .

Here is the complete verification. Multiplication by u gives $NR_j \subseteq R_{j-2}$. The only nonzero W -graded pieces are $\text{Gr}_{-1}^W V_2 = K$ and $\text{Gr}_1^W V_2 = Q$. The operator induced by N on each of these pieces is zero. The filtration induced by R on the first piece is concentrated in degree -1 , and on the second in degree $+1$, as proved in MRE14. Consequently, for every integer i and every $k \geq 0$,

$$N^k : \text{Gr}_{i+k}^R \text{Gr}_i^W V_2 \xrightarrow{\sim} \text{Gr}_{i-k}^R \text{Gr}_i^W V_2. \quad (\text{MRE29})$$

For $k = 0$, this is the identity. For $k > 0$, both source and target are zero, including when $i = -1$ or $i = 1$. For other i , the W -graded piece is itself zero. This proves every case of the relative-filtration identities.

In this two-dimensional extension uniqueness can also be proved directly. Any relative filtration R' induces the degree- -1 one-step filtration on K , and the degree- $+1$ one-step filtration on Q , because the induced nilpotent on each is zero and the relative graded isomorphisms force all other degrees to vanish. If $j < -1$, then $R'_j \cap K = 0$ and its image in Q is zero, so $R'_j = 0$. If $-1 \leq j < 1$, it contains K and its image in Q is zero, so $R'_j = K$. If $j \geq 1$, it contains K and surjects onto Q , so it equals V_2 . Thus $R' = R$, with every step determined.

This proves a relative filtration on the displayed original extension. Its additional eigenvalue shifts are MRE18 and MRE22. The proof assigns no independent geometric purity assertion to these constituents.

2.8 The original extension class is retained

Every linear section $s : Q \rightarrow V_2$ of p has the form

$$s(\bar{1}) = 1 + cu \quad (c \in \mathbb{C}). \quad (\text{MRE30})$$

For each such section,

$$Ns(\bar{1}) = u, \quad sN_Q(\bar{1}) = 0. \quad (\text{MRE31})$$

Hence no section intertwines the nilpotent operators: MRE9 is a non-split extension of nilpotent representations. The fixed nonzero map $\bar{N}$ of MRE23 records its extension coupling, and the Tate-typed isomorphism MRE26 retains that same coefficient-one coupling.

The arithmetic non-splitting is equally explicit. For every $a > 0$,

$$(A_{a,2}s - sA_{a,Q})(\bar{1}) = a^\rho(\log a)u. \quad (\text{MRE32})$$

The scalar is nonzero when $a \neq 1$, because $a^\rho \neq 0$ and $\log a \neq 0$. Thus no section intertwines even one fixed original arithmetic operator with $a \neq 1$, and no section intertwines the whole arithmetic family. The strict filtered exact sequence has therefore preserved the complete original arithmetic extension.

For the added diagonal operator alone, $s_0(\bar{1}) = 1$ is a filtered F_q -equivariant section. It is the only such section: equivariance for MRE30 forces $cq^{\rho-1/2} = cq^{\rho+1/2}$, and $q > 1$ forces $c = 0$. Equations MRE31–MRE32 calculate precisely how this additional diagonal splitting fails to split the original nilpotent and arithmetic actions. All three operator structures, their maps, and their difference are explicit on the original basis.

Their interaction also has an exact formula. Conjugating MRE10 gives

$$F_{q,2}A_{a,2}F_{q,2}^{-1} = a^\rho(I + q^{-1}(\log a)N), \quad F_{q,2}A_{a,2}F_{q,2}^{-1} - A_{a,2} = a^\rho(q^{-1} - 1)(\log a)N. \quad (\text{MRE33})$$

Every coefficient follows from $F_{q,2}NF_{q,2}^{-1} = q^{-1}N$. For $a \neq 1$, the last operator is nonzero, so the two retained actions do not commute. Equation MRE33 records their full interaction; no commuting-action assumption has been used in the filtered exact sequence or its twisted nilpotent maps.

2.9 Transport from the full original packet to its actual quotient

We give the exact downstream comparison, including the subobject, quotient, and all operators. In the original packet MW1 let $Z_{\geq 2} = \{\rho \in Z : m_\rho \geq 2\}$. The following construction is indexed by this subset and makes no assertion that it is nonempty. For each $\rho \in Z_{\geq 2}$ write $m = m_\rho$, $b = m - 2$, and retain the original $v_{\rho,r} = e_\rho(x - \rho)^r$. Define

$$\begin{aligned} J_\rho &= \text{span}\{v_{\rho,r} : 2 \leq r < m\}, \quad C_\rho = E_\rho/J_\rho, \quad \pi_\rho \left(\sum_{r=0}^{m-1} c_r v_{\rho,r} \right) = c_0 w_0 + c_1 w_1, \\ K_\rho &= \mathbb{C}w_1, \quad Q_\rho = C_\rho/K_\rho, \quad i_\rho(cw_1) = cw_1, \quad p_\rho(c_0 w_0 + c_1 w_1) = c_0[w_0], \\ \phi_\rho : C_\rho &\longrightarrow V_2, \quad \phi_\rho(c_0 w_0 + c_1 w_1) = c_0 + c_1 u. \end{aligned} \quad (\text{MRE34})$$

Here $w_r = \pi_\rho(v_{\rho,r})$, $r = 0, 1$, and the coordinate $u = x - \rho$ is fixed as in MRE7. Multiplication of original powers shows that $J_\rho = u_\rho^2 E_\rho$ is an ideal and that w_0 is the quotient unit with $w_1^2 = 0$. The map ϕ_ρ is an algebra isomorphism: its inverse sends $c_0 + c_1 u$ to $c_0 w_0 + c_1 w_1$, and the multiplication rules on the two indicated bases agree. The maps i_ρ, p_ρ give an exact sequence, since i_ρ is injective, p_ρ is surjective, and $\ker p_\rho = \text{im } i_\rho = K_\rho$.

The filtration on C_ρ is the quotient of the full original MW3 filtration. Its two degrees are retained explicitly:

$$\begin{aligned} \bar{M}_j C_\rho &= \pi_\rho(M_j E_\rho) = \begin{cases} 0 & j < b - 1, \\ K_\rho & b - 1 \leq j < b + 1, \\ C_\rho & j \geq b + 1, \end{cases} \quad \phi_\rho(\bar{M}_j C_\rho) = M_{j-b} V_2, \\ \bar{M}_j K_\rho &= K_\rho \cap \bar{M}_j C_\rho = \begin{cases} 0 & j < b - 1, \\ K_\rho & j \geq b - 1, \end{cases} \quad \bar{M}_j Q_\rho = p_\rho(\bar{M}_j C_\rho) = \begin{cases} 0 & j < b + 1, \\ Q_\rho & j \geq b + 1, \end{cases} \\ 0 &\longrightarrow \bar{M}_j K_\rho \xrightarrow{i_\rho} \bar{M}_j C_\rho \xrightarrow{p_\rho} \bar{M}_j Q_\rho \longrightarrow 0. \end{aligned} \quad (\text{MRE35})$$

Only the exponents zero and one survive π_ρ ; MW3 places them in degrees $m - 1 = b + 1$ and $m - 3 = b - 1$. This proves the first line and, by intersection and image, the second. The last line is zero in every position for $j < b - 1$; for $b - 1 \leq j < b + 1$, its only nonzero map is the identity on K_ρ ; for $j \geq b + 1$, it is the full exact sequence just proved. These three cases prove exactness and strictness. In terms of the separately centered degree-zero line filtrations C_j , the actual shifts are $\overline{M}_j K_\rho = C_{j-b+1} K_\rho$ and $\overline{M}_j Q_\rho = C_{j-b-1} Q_\rho$. The same coordinate calculation proves that $0 \rightarrow J_\rho \cap M_j E_\rho \rightarrow M_j E_\rho \xrightarrow{\pi_\rho} \overline{M}_j C_\rho \rightarrow 0$ is exact: its kernel is the displayed intersection and its image is the defined quotient step. Thus the map from the actual packet is strict before the subobject/quotient extension is formed.

All retained operators preserve J_ρ : the nilpotent raises exponents, each term of the original finite exponential raises exponents, and the added diagonals preserve every original basis line. Denote their induced operators on C_ρ by bars. Their entire formulas and their restriction and quotient values are

$$\begin{aligned} \overline{N}w_0 &= w_1, \quad \overline{N}w_1 = 0, \quad \overline{A}_a(c_0w_0 + c_1w_1) = a^\rho(c_0w_0 + (c_1 + c_0 \log a)w_1), \\ \overline{\Gamma}_q w_r &= q^{(b+1)/2-r} w_r, \quad \overline{F}_q w_r = q^{\rho+(b+1)/2-r} w_r \quad (r = 0, 1), \\ \begin{array}{c|ccc} & A_a & \Gamma_q & F_q \\ \hline K_\rho & a^\rho I & q^{(b-1)/2} I & q^{\rho+(b-1)/2} I \\ Q_\rho & a^\rho I & q^{(b+1)/2} I & q^{\rho+(b+1)/2} I \end{array} & \quad \text{(MRE36)} \\ \overline{A}_q &= \overline{F}_q \overline{\Gamma}_q^{-1} (I + (\log q) \overline{N}), \\ 2 \log_q |F_{q,K_\rho}| &= 2 \operatorname{Re} \rho + b - 1, \quad 2 \log_q |F_{q,Q_\rho}| = 2 \operatorname{Re} \rho + b + 1. \end{aligned}$$

In the full source exponential, terms with nilpotent power at least two have image in J_ρ ; applying π_ρ to each term therefore proves the first line with exactly its coefficient $a^\rho \log a$. The source diagonal exponents are $(m - 1)/2 - r = (b + 1)/2 - r$, proving the second line. Restriction to w_1 and passage to $[w_0]$ prove the table. For each $O = A_a, \Gamma_q, F_q$, applying to w_1 proves $\overline{O}i_\rho = i_\rho O_{K_\rho}$, and applying to both w_0, w_1 proves $p_\rho \overline{O} = O_{Q_\rho} p_\rho$. For the nilpotent these maps intertwine its zero restriction and quotient. The product $\overline{F}_q \overline{\Gamma}_q^{-1}$ is $q^\rho I$, proving reconstruction in the displayed order. The two real logarithms follow from $|q^{\rho+c}| = q^{\operatorname{Re} \rho + c}$ for real c .

For a precise filtered comparison of entire exact sequences, take the line $D_b = \mathbb{C}d_b$ concentrated in degree b , with $N_{D_b} = 0$, $A_{a,D_b} = I$, and $\Gamma_{q,D_b} = F_{q,D_b} = q^{b/2} I$. This is a specified complex filtered line for every integer $b = m - 2$; its definition does not assert a geometric realization. Give tensors sum filtrations, tensor-sum nilpotents, and tensor-product invertible operators. The comparison maps are

$$\begin{aligned} \theta_C : C_\rho &\xrightarrow{\sim} V_2 \otimes D_b, \quad w_0 \mapsto 1 \otimes d_b, \quad w_1 \mapsto u \otimes d_b, \\ \theta_K : K_\rho &\xrightarrow{\sim} K \otimes D_b, \quad w_1 \mapsto u \otimes d_b, \quad \theta_Q : Q_\rho \xrightarrow{\sim} Q \otimes D_b, \quad [w_0] \mapsto \overline{1} \otimes d_b, \\ \theta_C i_\rho &= (i \otimes I) \theta_K, \quad (p \otimes I) \theta_C = \theta_Q p_\rho, \\ \phi_\rho \overline{A}_a &= A_{a,2} \phi_\rho, \quad \phi_\rho \overline{\Gamma}_q = q^{b/2} \Gamma_{q,2} \phi_\rho, \quad \phi_\rho \overline{F}_q = q^{b/2} F_{q,2} \phi_\rho. \end{aligned} \quad \text{(MRE37)}$$

The inverse maps send the displayed target basis vectors to their displayed sources. The filtration step on each tensor is the original MRE13–MRE14 step at $j - b$, tensored with D_b . Equation MRE35 therefore proves strictness in both directions for each comparison map. The two diagram equations hold on w_1 and on w_0, w_1 , respectively, proving commutativity of the whole exact-sequence comparison. The last three equations follow by applying both sides to w_0, w_1 , using MRE17 and MRE36. The same calculation intertwines the nilpotents with coefficient one. Thus all maps in the sequence are intertwined by the specified θ 's for N, A, Γ, F . For $m = 2$, $b = 0$ and these are exactly the fixed length-two maps. For $m > 2$, ϕ_ρ with separately centered target filtration is filtration preserving because $j - b \leq j$, but non-strict: at $j = 1$ its image of $\overline{M}_1 C_\rho$ omits 1, whereas $M_1 V_2 = V_2$. The scalar $q^{b/2}$ in both added operators and the degree shift b are retained in MRE37.

The actual nilpotent link is

$$\begin{aligned}\overline{N} &= i_\rho \eta_\rho p_\rho, & \eta_\rho : Q_\rho &\xrightarrow{\sim} K_\rho, & \eta_\rho([w_0]) &= w_1, \\ \nu_\rho : Q_\rho \otimes T_q &\xrightarrow{\sim} K_\rho, & \nu_\rho([w_0] \otimes t) &= w_1, \\ F_{q,K_\rho} \eta_\rho &= q^{-1} \eta_\rho F_{q,Q_\rho}, & \Gamma_{q,K_\rho} \eta_\rho &= q^{-1} \eta_\rho \Gamma_{q,Q_\rho}, & A_{a,K_\rho} \eta_\rho &= \eta_\rho A_{a,Q_\rho}.\end{aligned}\tag{MRE38}$$

Here T_q is exactly MRE25, with degree -2 , both added scalars q^{-1} , and arithmetic scalar one. Applying the factorization to $c_0 w_0 + c_1 w_1$ gives $c_0 w_1$ on both sides. In ν_ρ , the domain and target both have degree $(b+1) - 2 = b-1$, added F -eigenvalue $q^{\rho+(b+1)/2} q^{-1} = q^{\rho+(b-1)/2}$, and the analogous Γ -eigenvalue with ρ omitted. Their original arithmetic eigenvalues are both a^ρ . Its inverse is $w_1 \mapsto [w_0] \otimes t$. These computations prove strict filtered equivariance, its inverse, and all three identities in the last line. The comparison with MRE26 sends $(\overline{1} \otimes d_b) \otimes t$ to $(\overline{1} \otimes t) \otimes d_b$ and then to $u \otimes d_b$; the specified tensor rearrangement has coefficient one, so this is exactly $\theta_K \nu_\rho$.

The whole quotient has the corresponding strict map

$$\begin{aligned}\mathcal{N}_\rho : C_\rho \otimes T_q &\longrightarrow C_\rho, & z \otimes t &\longmapsto \overline{N}z, \\ \overline{N}(\overline{M}_{j+2} C_\rho) &= K_\rho \cap \overline{M}_j C_\rho, & \pi_\rho \mathcal{N}_{E_\rho} &= \mathcal{N}_\rho(\pi_\rho \otimes I), & \mathcal{N}_{E_\rho}(v \otimes t) &= N_\rho v.\end{aligned}\tag{MRE39}$$

For $j < b-1$, the domain step in the middle equality is at most K_ρ , which $\overline{N}$ kills, and the right side is zero. For $j \geq b-1$, that domain step is C_ρ , its image is K_ρ , and the right side is K_ρ . This proves strictness at every integer. The quotient diagonal entries give $\overline{F}_q \overline{N} \overline{F}_q^{-1} = q^{-1} \overline{N}$ and the identical relation for $\overline{\Gamma}_q$; the full arithmetic formula gives $\overline{A}_a \overline{N} = \overline{N} \overline{A}_a$. These prove equivariance with the three operators on the Tate line. The last equality follows on $v \otimes t$ from the already proved identity $\pi_\rho N_\rho = \overline{N} \pi_\rho$.

The relative-filtration comparison shifts both filtrations of MRE28–MRE29. Define

$$\begin{aligned}W_i^{(\rho)} C_\rho &= \phi_\rho^{-1}(W_{i-b} V_2), & R_j^{(\rho)} C_\rho &= \phi_\rho^{-1}(R_{j-b} V_2), & W^{(\rho)} &= R^{(\rho)} = \overline{M}, \\ \overline{N}^k : \mathrm{Gr}_{i+k}^{R^{(\rho)}} \mathrm{Gr}_i^{W^{(\rho)}} C_\rho &\xrightarrow{\sim} \mathrm{Gr}_{i-k}^{R^{(\rho)}} \mathrm{Gr}_i^{W^{(\rho)}} C_\rho & (i \in \mathbb{Z}, k \geq 0).\end{aligned}\tag{MRE40}$$

Multiplication by the local coordinate lowers the displayed degree by two. The only nonzero $W^{(\rho)}$ -graded spaces are K_ρ in degree $b-1$ and Q_ρ in degree $b+1$. The induced nilpotents on them are zero, and the induced $R^{(\rho)}$ -filtrations are concentrated in their respective degrees by MRE35. Thus for $k=0$ the displayed map is the identity, and for $k>0$ both spaces are zero. The comparison ϕ_ρ identifies each iterated graded space with $\mathrm{Gr}_{i-b\pm k}^R \mathrm{Gr}_{i-b}^W V_2$; hence these are exactly the MRE29 maps with i replaced by $i-b$. Any other relative filtration must induce degree $b-1$ on K_ρ and degree $b+1$ on Q_ρ , because for a zero nilpotent the relative isomorphisms force all other graded pieces to vanish. Below $b-1$ its intersection with K_ρ and image in Q_ρ are zero, so the step is zero. Between $b-1$ and $b+1$ it contains K_ρ with zero quotient image, so it equals K_ρ . From $b+1$ onward it contains K_ρ and surjects onto Q_ρ , so it equals C_ρ . This proves uniqueness with all induced degrees intact. The displayed $W^{(\rho)}$ is also the quotient of the specified full-block filtration $W_j^{\mathrm{full}} E_\rho = M_j E_\rho$, since its quotient steps are MRE35. This is the specified filtration on the original coefficient packet, with its exact quotient map. For the separately specified one-step filtration on C_ρ concentrated in degree b , MRE5–MRE6 likewise identify $\overline{M}$ as its relative filtration: the only distance-one map sends $[w_0]$ to $[w_1]$ with coefficient one, distance zero gives the identity, and every distance at least two has both graded spaces zero.

The extension discrepancies also survive without any factor change. Every linear section $s_c : Q_\rho \rightarrow C_\rho$ of p_ρ sends $[w_0]$ to $w_0 + c w_1$. Direct substitution gives

$$\begin{aligned}(\overline{N} s_c - s_c N_{Q_\rho})([w_0]) &= w_1, & (\overline{A}_a s_c - s_c A_{a,Q_\rho})([w_0]) &= a^\rho (\log a) w_1, \\ (\overline{F}_q s_c - s_c F_{q,Q_\rho})([w_0]) &= c(q^{\rho+(b-1)/2} - q^{\rho+(b+1)/2}) w_1, \\ \overline{F}_q \overline{A}_a \overline{F}_q^{-1} - \overline{A}_a &= a^\rho (q^{-1} - 1) (\log a) \overline{N}.\end{aligned}\tag{MRE41}$$

The first two differences are independent of c ; the second is nonzero for $a \neq 1$. Thus no section splits either the nilpotent or an original arithmetic operator with $a \neq 1$. The third difference vanishes only for $c = 0$, since its parenthesized factor is $q^{\rho+(b-1)/2}(1-q) \neq 0$. The section s_0 preserves the filtration because its only source degree is $b+1$, when the target is already the full space. It therefore gives exactly the added diagonal splitting. Conjugating the full arithmetic formula in MRE36 by $\bar{F}_q$ and using its q^{-1} nilpotent relation proves the last line with its unchanged scalar. These are the transported MRE30–MRE33 calculations on the actual quotient.

Finally the transport is compatible with every independent slot and with the combined operator of MW9a. For an admissible odd prime power q coprime to n , extend π_ρ linearly to the coefficient algebra and put $\bar{T}_{a,q} = \bar{A}_a \Psi_q$. On the fixed bases,

$$\begin{aligned} \bar{T}_{a,q}(w_r t^i) &= (-1)^{\lfloor qi/n \rfloor} a^\rho \sum_{\ell=0}^{1-r} \frac{(\log a)^\ell}{\ell!} w_{r+\ell} t^{qi \bmod n}, \quad r = 0, 1, \\ \pi_\rho T_{a,q} &= \bar{T}_{a,q} \pi_\rho, \quad \bigotimes_{\nu} \theta_{C, \rho_\nu} : \bigotimes_{\nu} C_{\rho_\nu} \xrightarrow{\sim} \left(\bigotimes_{\nu} V_{2, \rho_\nu} \right) \otimes \left(\bigotimes_{\nu} D_{b_\nu} \right), \\ b_\nu &= m_{\rho_\nu} - 2, \quad \text{degree translation } \sum_{\nu} b_\nu, \quad \text{added } F, \Gamma \text{ factor } q^{\frac{1}{2} \sum_{\nu} b_\nu}. \end{aligned} \tag{MRE42}$$

Linearity of π_ρ preserves the displayed sign, and applying it to the full source finite Taylor sum keeps exactly the terms shown. This proves both the first line and the intertwining identity. On a face the map retains the mask and sends each coefficient to its quotient; a vanished coefficient remains the zero of that same present slot. Zero insertion commutes with π_ρ , since it is linear. For the tensor comparison, send each displayed original tensor basis vector to $(\bigotimes_{\nu} u^{r_\nu}) \otimes (\bigotimes_{\nu} d_{b_\nu})$, with the inverse prescribed by the same basis pairs. Every filtration degree is translated by the sum $\sum_{\nu} b_\nu$, and both added diagonal factors multiply to $q^{\frac{1}{2} \sum_{\nu} b_\nu}$. This proves strictness and operator compatibility of this full tensor isomorphism. On coefficient eigenspaces with eigenvalues ω_ν , the original arithmetic graded eigenvalue is $a^{\sum_{\nu} \rho_\nu}$, the coefficient eigenvalue is $\prod_{\nu} \omega_\nu$, and their combined eigenvalue is $(\prod_{\nu} \omega_\nu) a^{\sum_{\nu} \rho_\nu}$, exactly as in MW16b. Positive nilpotent powers lower total degree by two per power, so none changes these graded eigenvalues; every such power is still retained in the exact tensor formula MW16a.

Proved control and continuing scope. The calculation constructs the strict filtered exact sequence on the length-two arithmetic extension, its exact shifted added eigenvalues, the Tate-typed nilpotent isomorphism, and its relative filtration. Every original arithmetic scalar and nilpotent coefficient survives. These results supply this local extension control within the program. MRE34–MRE42 supply its strict comparison with every specified actual length-two packet quotient, including the original induced degree translation, both added diagonal factors, and the full arithmetic and coefficient operators. The arbitrary parameter calculation MRE7–MRE33 makes no zero-multiplicity assertion. They do not determine $2 \operatorname{Re} \rho$, establish a pure lisse source, or supply the analytic upper estimate used in Weil II 1.8.1. Those numerical and geometric controls remain separate constructions to be carried out on the original arithmetic sources.